

Министерство науки и высшего образования Российской Федерации
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ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ (НИ ТГУ)
Институт прикладной математики и информатики

ОТЧЕТ ПО ЛАБОРАТОРНОЙ РАБОТЕ
по дисциплине «Системы управления базами данных»

Символьные и числовые функции

Бриц Владислав Олегович

Специалитет, 10.05.01,
Направленность «Компьютерная Безопасность»

Руководитель работы
Старший преподаватель кафедры
теоретических основ информатики

_____ Е.Е. Мокина

подпись

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Автор работы
студент группы № 932025

_____ В.О. Бриц

подпись

«17» ноября 2022 г.

1. Найти модули чисел -100, 0, 9

SELECT ABS(-100), ABS(0), ABS(9) FROM DUAL;

The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying the SQL query: `SELECT ABS(-100), ABS(0), ABS(9) FROM DUAL;`. Below the query editor, the 'Query Result' window is open, showing the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0,004 seconds'. The results are displayed in a table with three columns: `ABS(-100)`, `ABS(0)`, and `ABS(9)`. The first row contains the values 1, 100, and 9.

	ABS(-100)	ABS(0)	ABS(9)
1	100	0	9

2. Найти приближенные целые числа от чисел -1.2 и 9.8

SELECT CEIL(-1.2), FLOOR(-1.2), CEIL(9.8), FLOOR(9.8) FROM DUAL;

The screenshot shows the SQL Developer interface. The 'Worksheet' tab is active, displaying the SQL query: `SELECT CEIL(-1.2), FLOOR(-1.2), CEIL(9.8), FLOOR(9.8) FROM DUAL;`. Below the query editor, the 'Query Result' window is open, showing the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0,003 seconds'. The results are displayed in a table with four columns: `CEIL(-1.2)`, `FLOOR(-1.2)`, `CEIL(9.8)`, and `FLOOR(9.8)`. The first row contains the values 1, -1, -2, 10, and 9.

	CEIL(-1.2)	FLOOR(-1.2)	CEIL(9.8)	FLOOR(9.8)
1	-1	-2	10	9

3. Найти first_name, last_name и символ по ASCII значению первой буквы last_name, у которых first_name совпадает с David

SELECT first_name, last_name, CHR(ASCII(last_name)) FROM HR.employees WHERE first_name = 'David';

The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT first_name, last_name, CHR(ASCII(last_name)) FROM HR.employees  
WHERE first_name = 'David';
```

Below the query editor, the 'Query Result' window shows the results of the query. It indicates that all rows were fetched in 0.004 seconds. The results are displayed in a table with three columns: FIRST_NAME, LAST_NAME, and CHR(ASCII(LAST_NAME)).

	FIRST_NAME	LAST_NAME	CHR(ASCII(LAST_NAME))
1	David	Austin	A
2	David	Bernstein	B
3	David	Lee	L

4. Найти значения выражений $\cos(2\pi)$, $\sin(\pi/2)$, $\text{tg}(\pi/4)$

SELECT ROUND(COS(2*3.141593)), ROUND(SIN(3.141593/2)), ROUND(TAN(3.141593/4)) FROM DUAL;

The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT ROUND(COS(2*3.141593)), ROUND(SIN(3.141593/2)), ROUND(TAN(3.141593/4))  
FROM DUAL;
```

Below the query editor, the 'Query Result' window shows the results of the query. It indicates that all rows were fetched in 0.002 seconds. The results are displayed in a table with three columns: ROUND(COS(2*3.141593)), ROUND(SIN(3.141593/2)), and ROUND(TAN(3.141593/4)).

	ROUND(COS(2*3.141593))	ROUND(SIN(3.141593/2))	ROUND(TAN(3.141593/4))
1	1	1	1

5. Найти значения выражений cosh(1.5), sinh(2), tgh(5)

SELECT COSH(1.5), SINH(2), TANH(5) FROM DUAL;

The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the query: `SELECT COSH(1.5), SINH(2), TANH(5) FROM DUAL;`. Below the query editor, the 'Query Result' tab shows the execution status: 'All Rows Fetched: 1 in 0,002 seconds'. The result is displayed in a table with three columns: COSH(1.5), SINH(2), and TANH(5). The first row contains the values: 2,35240961524324732576766796544164417018, 3,62686040784701876766821398280126170488, and 0,999909204262595131210990447534473021089.

COSH(1.5)	SINH(2)	TANH(5)
2,35240961524324732576766796544164417018	3,62686040784701876766821398280126170488	0,999909204262595131210990447534473021089

6. Вывести фамилию, должность в нижнем регистре и номер отдела для всех сотрудников у кого зарплата за полугодие меньше 15000.

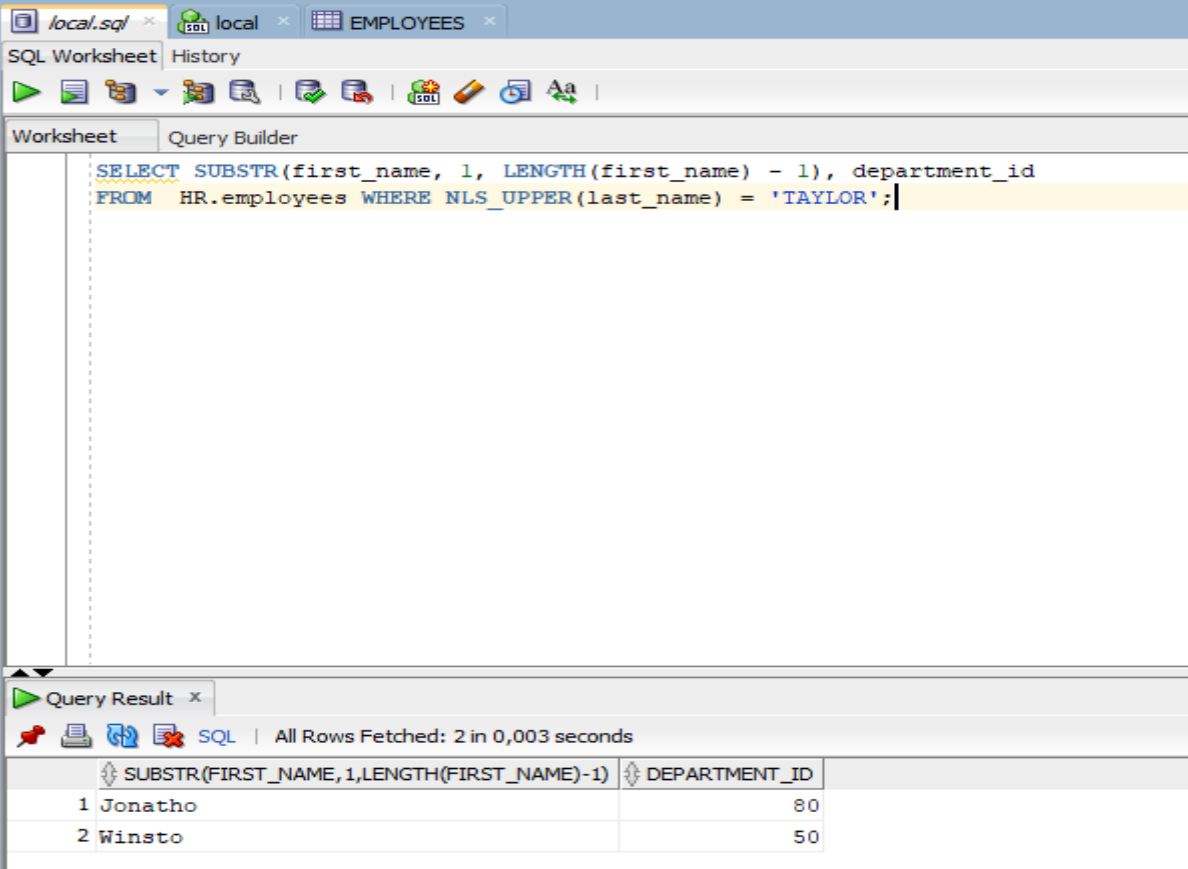
SELECT last_name, NLS_LOWER(JOB_ID), department_id FROM HR.employees WHERE salary*6 < 15000

The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the query: `SELECT last_name, NLS_LOWER(JOB_ID), department_id FROM HR.employees WHERE salary*6 < 15000;`. Below the query editor, the 'Query Result' tab shows the execution status: 'All Rows Fetched: 5 in 0,003 seconds'. The result is displayed in a table with three columns: LAST_NAME, NLS_LOWER(JOB_ID), and DEPARTMENT_ID. The first five rows show employees with the job ID 'st_clerk' in department 50.

LAST_NAME	NLS_LOWER(JOB_ID)	DEPARTMENT_ID
1 Landry	st_clerk	50
2 Markle	st_clerk	50
3 Olson	st_clerk	50
4 Gee	st_clerk	50
5 Philtanker	st_clerk	50

7. Вывод имени без последнего символа и номера отдела для сотрудника по фамилии «TAYLOR».

```
SELECT SUBSTR(first_name, 1, LENGTH(first_name) - 1), department_id FROM HR.employees WHERE NLS_UPPER(last_name) = 'TAYLOR';
```



The screenshot shows a SQL Worksheet interface with a query entered in the Query Builder tab. The query is: `SELECT SUBSTR(first_name, 1, LENGTH(first_name) - 1), department_id FROM HR.employees WHERE NLS_UPPER(last_name) = 'TAYLOR';`. The results are displayed in the Query Result tab, showing two rows of data.

	SUBSTR(FIRST_NAME,1,LENGTH(FIRST_NAME)-1)	DEPARTMENT_ID
1	Jonatho	80
2	Winsto	50

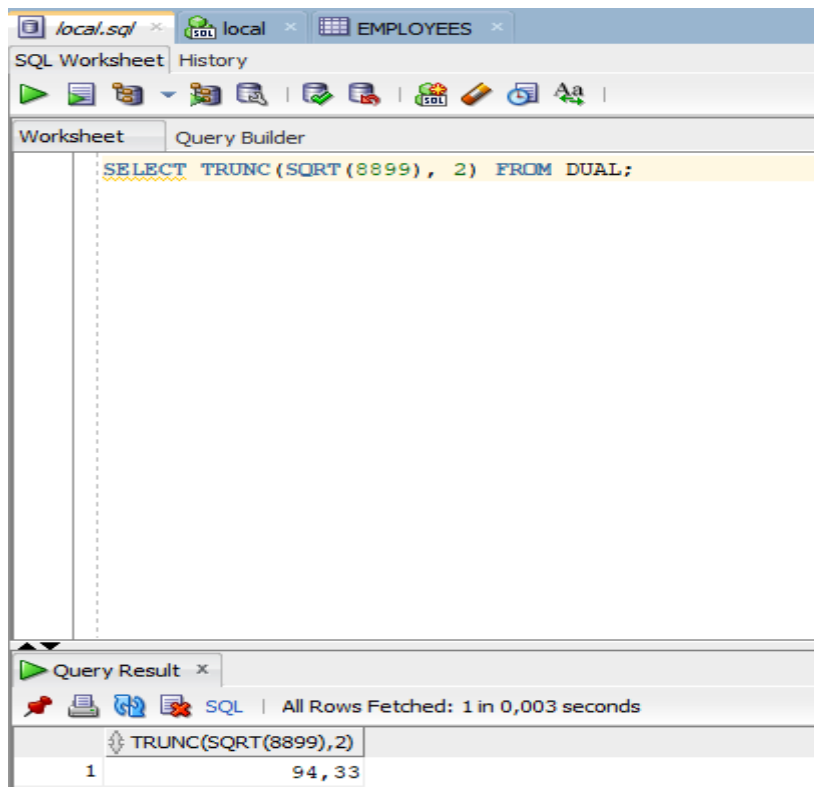
8. Вывод имени в нижнем регистре, зарплаты за год, округленной до сотен, для всех сотрудников, кроме сотрудника с фамилией «biri».

```
SELECT NLS_LOWER(first_name), ROUND(salary*12, -2) FROM HR.employees WHERE NLS_UPPER(last_name) != 'BIRI';
```

local.sql	
local	
EMPLOYEES	
SQL Worksheet	
History	
Worksheet	
Query Builder	
<pre>SELECT NLS_LOWER(first_name), ROUND(salary*12, -2) FROM HR.employees WHERE NLS_UPPER(last_name) != 'BIRI';</pre>	
Query Result	
SQL All Rows Fetched: 107 in 0,01 seconds	
NLS_LOWER(FIRST_NAME)	ROUND(SALARY*12,-2)
1 donald	31200
2 douglas	31200
3 jennifer	52800
4 michael	156000
5 pat	72000
6 susan	78000
7 hermann	120000
8 shelly	144100
9 william	99600
10 steven	288000

9. Найти корень из 8899, усеченный до сотых.

SELECT TRUNC(SQRT(8899), 2) FROM DUAL;



10. Вывести фамилию с длиной 5 символов и комиссионные для всех служащих с зарплатой менее 7000.

```
SELECT last_name, salary*commission_pct FROM HR.employees WHERE (LENGTH(last_name) = 5 and salary < 7000);
```

local.sql		
local		
EMPLOYEES		
SQL Worksheet		
History		
Worksheet		
Query Builder		
<pre>SELECT last_name, salary*commission_pct FROM HR.employees WHERE (LENGTH(last_name) = 5 and salary < 7000);</pre>		
Query Result		
All Rows Fetched: 15 in 0,003 seconds		
	LAST_NAME	SALARY*COMMISSION_PCT
1	Grant	(null)
2	Ernst	(null)
3	Baida	(null)
4	Nayer	(null)
5	Olson	(null)
6	Patel	(null)
7	Matos	(null)
8	Banda	620
9	Kumar	610
10	Geoni	(null)
11	Chung	(null)
12	Dilly	(null)
13	Gates	(null)
14	Jones	(null)
15	Walsh	(null)

11. Найти значения e^3 , $\ln(4)$, $\ln(e)$.

SELECT EXP(3), LN(4), LN(EXP(1)) FROM DUAL;

local.sql x local x EMPLOYEES x

SQL Worksheet History

Worksheet Query Builder

```
SELECT EXP(3), LN(4), LN(EXP(1)) FROM DUAL;
```

Query Result x

SQL | All Rows Fetched: 1 in 0,002 seconds

	EXP(3)	LN(4)	LN(EXP(1))
1	20,08553692318766774092852965458171789706	1,38629436111989061883446424291635313614	1

12. Вывести email, длина которого равна 5, и этот же email, но у которого будет первая буква в верхнем регистре, а остальные в нижнем.

SELECT email, NLS_INITCAP(email) "EMAIL*" FROM HR.employees WHERE LENGTH(email) = 5;

local.sql x local x EMPLOYEES x

SQL Worksheet History

Worksheet Query Builder

```
SELECT email, NLS_INITCAP(email) "EMAIL*" FROM HR.employees WHERE LENGTH(email) = 5;
```

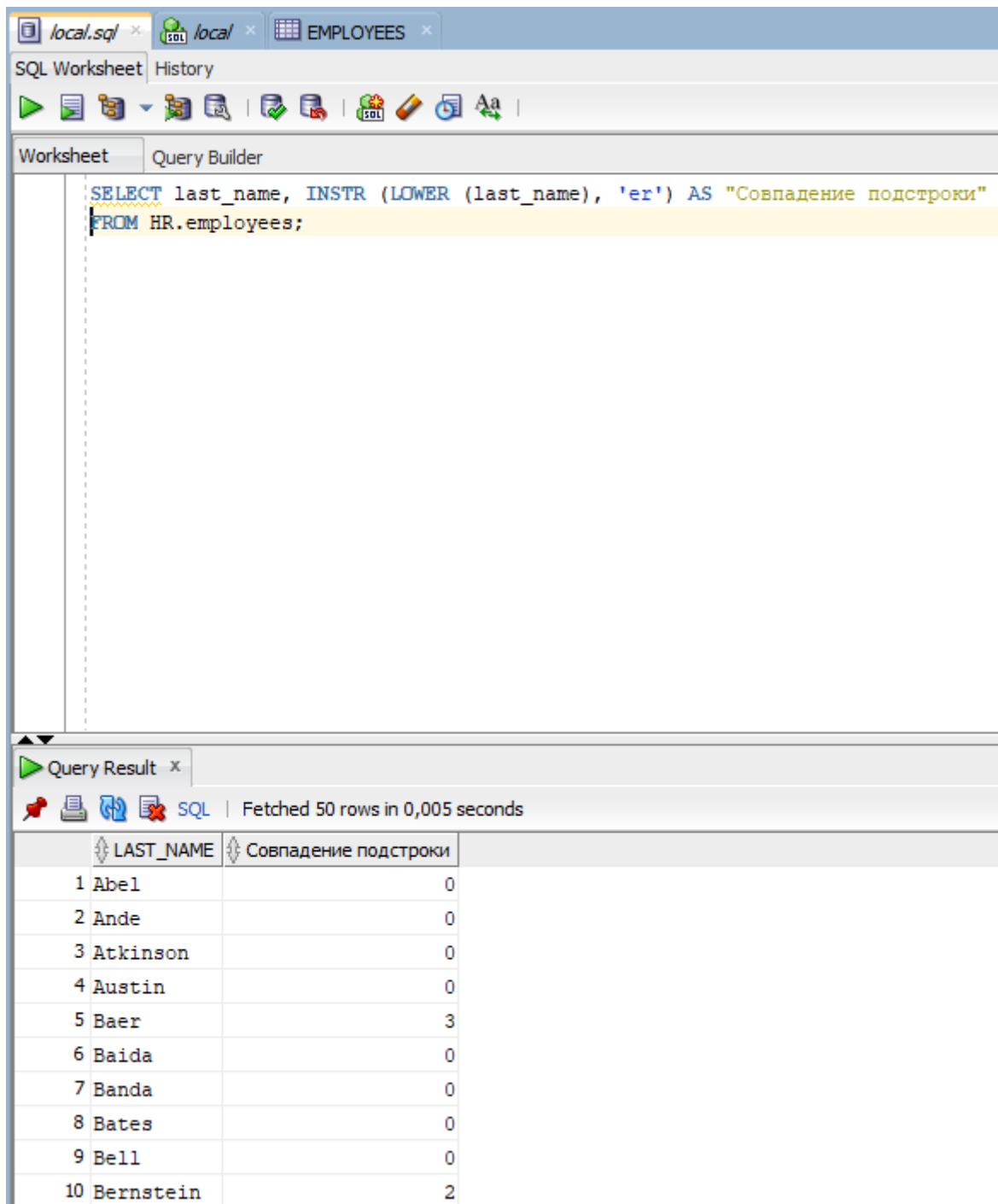
Query Result x

SQL | All Rows Fetched: 13 in 0,003 seconds

	EMAIL	EMAIL*
1	ABULL	Abull
2	AKHOO	Akhoo
3	EABEL	Eabel
4	HBAER	Hbaer
5	JCHEN	Jchen
6	JKING	Jking
7	LOZER	Lozer
8	LPOPP	Lpopp
9	PHALL	Phall
10	SANDE	Sande
11	SBELL	Sbell
12	SKING	Sking
13	TRAJS	Trajs

13. Вывод фамилий сотрудников и нахождение позиции совпадения первого символа в подстроке 'er' с фамилией из таблицы Служащие.

```
SELECT last_name, INSTR(LOWER(last_name), 'er') AS "Совпадение подстроки" FROM HR.employees;
```



The screenshot shows the SQL Developer interface. The top pane displays the following SQL query:

```
SELECT last_name, INSTR (LOWER (last_name), 'er') AS "Совпадение подстроки"
FROM HR.employees;
```

The bottom pane shows the query results in a table with two columns: LAST_NAME and Совпадение подстроки. The results are as follows:

	LAST_NAME	Совпадение подстроки
1	Abel	0
2	Ande	0
3	Atkinson	0
4	Austin	0
5	Baer	3
6	Baida	0
7	Banda	0
8	Bates	0
9	Bell	0
10	Bernstein	2

14. Вывод имен сотрудников и их длины в байтах из таблицы Служащие.

```
SELECT first_name, LENGTH(first_name) AS "Length example" FROM HR.employees;
```

local.sql x local x EMPLOYEES x

SQL Worksheet History

Worksheet Query Builder

```
SELECT first_name, LENGTH(first_name) AS "Length example" FROM HR.employees;
```

Query Result x

SQL | Fetched 50 rows in 0,005 seconds

	FIRST_NAME	Length example
1	Ellen	5
2	Sundar	6
3	Mozhe	5
4	David	5
5	Hermann	7
6	Shelli	6
7	Amit	4
8	Elizabeth	9
9	Sarah	5
10	David	5
11	Laura	5
12	Harrison	8
13	Alexis	6
14	Anthony	7
15	Gerald	6

15. Найти значения $\log_2(3)$, $\log_{10}(10)$, $\log_9(1)$.

SELECT LOG(2, 3), LOG(10, 10), LOG(9, 1) FROM DUAL;

The screenshot shows the SQL Developer interface. The top pane (Worksheet) contains the following SQL query:

```
SELECT LOG(2,3), LOG(10,10), LOG(9,1) FROM DUAL;
```

The bottom pane (Query Result) shows the results of the query. It indicates that all rows were fetched in 0.002 seconds. The results are displayed in a table with three columns: LOG(2,3), LOG(10,10), and LOG(9,1).

LOG(2,3)	LOG(10,10)	LOG(9,1)
1,58496250072115618145373894394781650876	1	0

16. Вычисление остатка от деления заработной платы на комиссионные для всех служащих с зарплатой выше 1000.

```
SELECT last_name, MOD(salary, commision_pct) AS "MOD example" FROM HR.employees
WHERE salary > 1000;
```

The screenshot shows the SQL Developer interface. The top pane (Worksheet) contains the following SQL query:

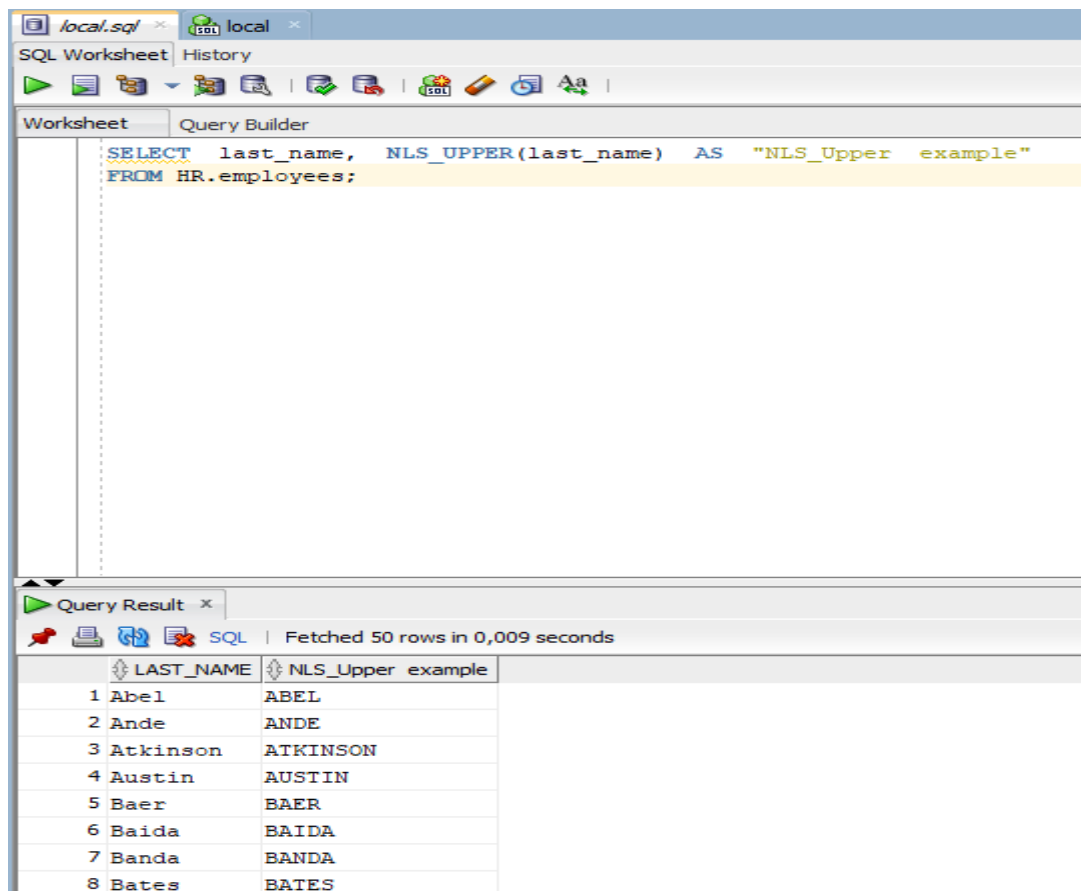
```
SELECT last_name, MOD(salary, commission_pct) AS "MOD example"
FROM HR.employees WHERE salary > 1000;
```

The bottom pane (Query Result) shows the results of the query. It indicates that 100 rows were fetched in 0.005 seconds. The results are displayed in a table with two columns: LAST_NAME and MOD example.

LAST_NAME	MOD example
52 Davies	(null)
53 Matos	(null)
54 Vargas	(null)
55 Russell	0
56 Partners	0
57 Errazuriz	0
58 Cambrault	0,2
59 Zlotkey	0
60 Tucker	0,1
61 Bernstein	0

17. Преобразование в верхний регистр фамилий сотрудников.

```
SELECT last_name, NLS_UPPER(last_name) AS "NLS_Upper example" FROM HR.employees;
```



The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT last_name, NLS_UPPER(last_name) AS "NLS_Upper example"
FROM HR.employees;
```

Below the query editor, the 'Query Result' window shows the execution results. It indicates that 50 rows were fetched in 0.009 seconds. The results are displayed in a table with two columns: LAST_NAME and NLS_Upper example.

	LAST_NAME	NLS_Upper example
1	Abel	ABEL
2	Ande	ANDE
3	Atkinson	ATKINSON
4	Austin	AUSTIN
5	Baer	BAER
6	Baida	BAIDA
7	Banda	BANDA
8	Bates	BATES

18. Вывести фамилии и имена сотрудников и отсортировать записи по алфавиту по именам.

```
SELECT first_name, last_name FROM HR.employees ORDER BY NLSSORT(first_name, 'NLS_sort =
XDanish');
```

The screenshot shows an SQL Worksheet interface with a tab labeled 'EMPLOYEES'. The query editor contains the following SQL statement:

```
SELECT first_name, last_name FROM HR.employees
ORDER BY NLSSORT(first_name, 'NLS_SORT = XDanish');
```

Below the query editor, the 'Query Result' pane displays the results of the query. It indicates that 50 rows were fetched in 0.003 seconds. The results are shown in a table with two columns: FIRST_NAME and LAST_NAME.

	FIRST_NAME	LAST_NAME
1	Adam	Fripp
2	Alana	Walsh
3	Alberto	Errazuriz
4	Alexander	Khoo
5	Alexander	Hunold
6	Alexis	Bull
7	Allan	McEwen
8	Alyssa	Hutton
9	Amit	Banda
10	Anthony	Cabrio
11	Britney	Everett
12	Bruce	Ernst
13	Charles	Johnson
14	Christopher	Olsen
15	Clara	Vishney

19. Замена в фамилиях сотрудников строки символов 'an' на строку 'bb'.

SELECT last_name, REPLACE(LOWER(last_name), 'an', 'bb') AS "Replace example" FROM HR.employees;

The screenshot shows an SQL IDE interface. At the top, there are tabs for 'local.sql' and 'local'. Below the tabs is a toolbar with various icons. The main area is divided into 'Worksheet' and 'Query Builder' tabs. The 'Worksheet' tab is active, displaying the following SQL query:

```
SELECT last_name, REPLACE (LOWER (last_name), 'an', 'bb')
AS "Replace example" FROM HR.employees;
```

Below the query editor, the 'Query Result' tab is active, showing the results of the query. The results are displayed in a table with two columns: 'LAST_NAME' and 'Replace example'. The table contains 9 rows of data:

	LAST_NAME	Replace example
1	Abel	abel
2	Ande	bbde
3	Atkinson	atkinson
4	Austin	austin
5	Baer	baer
6	Baida	baida
7	Banda	bbbda
8	Bates	bates
9	Bell	bell

20. Вывод значения 45.953 с округлением до сотых, до целых и с округлением до десятков.

```
SELECT ROUND(45.953, 2), ROUND(45.953, 0), ROUND(45.953, -1) FROM DUAL;
```

local.sql local EMPLOYEES

SQL Worksheet History

Worksheet Query Builder

```
SELECT ROUND(45.953,2), ROUND(45.953,0), ROUND(45.953,-1) FROM DUAL;
```

Query Result x

SQL | All Rows Fetched: 1 in 0,002 seconds

	ROUND(45.953,2)	ROUND(45.953,0)	ROUND(45.953,-1)
1	45,95	46	50

21. Вычислить значения функции знака в точках -1, 0, 1.

SELECT SIGN(-1), SIGN(0), SIGN(1) FROM DUAL;

local.sql local EMPLOYEES

SQL Worksheet History

Worksheet Query Builder

```
SELECT SIGN(-1), SIGN(0), SIGN(1) FROM DUAL;
```

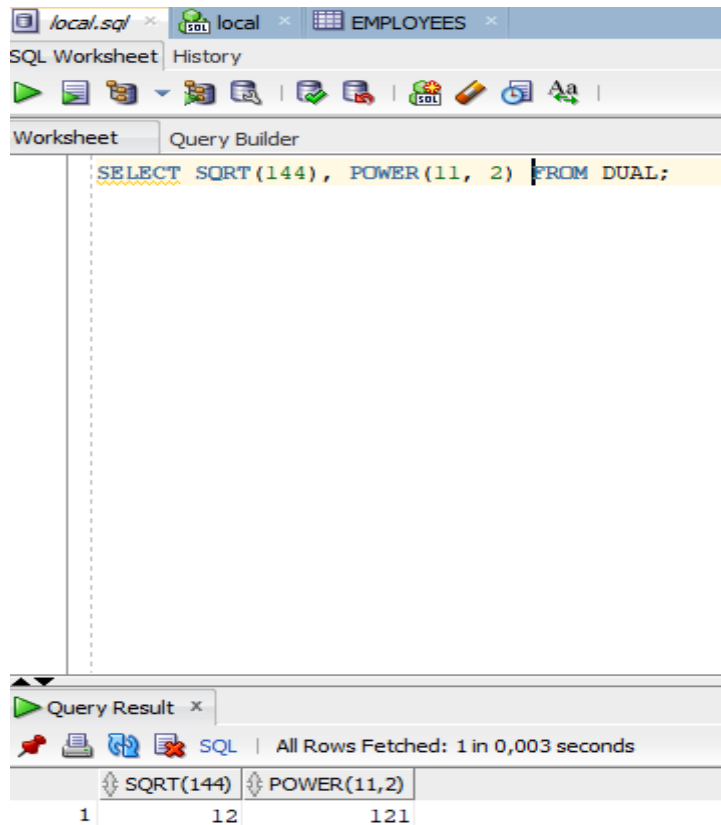
Query Result x

SQL | All Rows Fetched: 1 in 0,002 seconds

	SIGN(-1)	SIGN(0)	SIGN(1)
1	-1	0	1

22. Вычислить значения $\sqrt{144}$ и 11^2 .

SELECT SQRT(144), POWER(11, 2) FROM DUAL;



23. Вывести адреса улиц без индексов.

SELECT street_address, SUBSTR(street_address, INSTR(street_address, ' ') "STREET_ADDRESS*" FROM HR.locations WHERE (CHR(ASCII(street_address)) >= '0' and CHR(ASCII(street_address)) <= '9');

The screenshot shows an SQL IDE with a query window and a results window. The query window contains the following SQL statement:

```
SELECT street_address, SUBSTR(street_address, INSTR(street_address, ' '))
"STREET_ADDRESS*" FROM HR.locations WHERE (CHR(ASCII(street_address)) >= '0'
and CHR(ASCII(street_address)) <= '9');
```

The results window shows the following data:

	STREET_ADDRESS	STREET_ADDRESS*
1	1297 Via Cola di Rie	Via Cola di Rie
2	93091 Calle della Testa	Calle della Testa
3	2017 Shinjuku-ku	Shinjuku-ku
4	9450 Kamiya-cho	Kamiya-cho
5	2014 Jabberwocky Rd	Jabberwocky Rd
6	2011 Interiors Blvd	Interiors Blvd
7	2007 Zagora St	Zagora St
8	2004 Charade Rd	Charade Rd
9	147 Spadina Ave	Spadina Ave
10	6092 Boxwood St	Boxwood St
11	40-5-12 Laogianggen	Laogianggen
12	1298 Vileparle (E)	Vileparle (E)
13	12-98 Victoria Street	Victoria Street

24. Преобразование в именах сотрудников символов 'а' на символ 'о'.

```
SELECT first_name, TRANSLATE(LOWER(first_name, 'а', 'о') AS TRANSLATE_EXAMPLE FROM
HR.employees;
```

The screenshot shows a SQL Worksheet interface with a query editor and a results pane. The query in the editor is:

```
SELECT first_name, TRANSLATE (LOWER (first_name), 'a', 'o') AS TRANSLATE_EXAMPLE
FROM HR.employees;
```

The results pane shows the output of the query, displaying two columns: FIRST_NAME and TRANSLATE_EXAMPLE. The results are as follows:

	FIRST_NAME	TRANSLATE_EXAMPLE
4	David	dovid
5	Hermann	hermonn
6	Shelli	shelli
7	Amit	omit
8	Elizabeth	elizobeth
9	Sarah	soroh
10	David	dovid
11	Laura	louro
12	Harrison	horrison
13	Alexis	olexis
14	Anthony	onthony
15	Gerald	gerold
16	Nanette	nonette

25. Вывод значения 45.953 с усечением до сотых, до целых и до десятков.

SELECT TRUNC(45.953, 2), TRUNC(45.953, 0), TRUNC(45.953, -1) FROM DUAL;

local.sql x local x EMPLOYEES x

SQL Worksheet History

Worksheet Query Builder

```
SELECT TRUNC(45.953,2), TRUNC(45.953,0), TRUNC(45.953,-1) FROM DUAL;
```

Query Result x

SQL | All Rows Fetched: 1 in 0,002 seconds

	TRUNC(45.953,2)	TRUNC(45.953,0)	TRUNC(45.953,-1)
1	45,95	45	40